Лабораторная №1

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Задание 1

Код программы

```
import matplotlib.pyplot as plt
import os

FILE_NAME = "001.dat"

FILE_PATH = os.path.join(
    r"C:\Users\user\Downloads\dead_moroz\dead_moroz",
    FILE_NAME
)

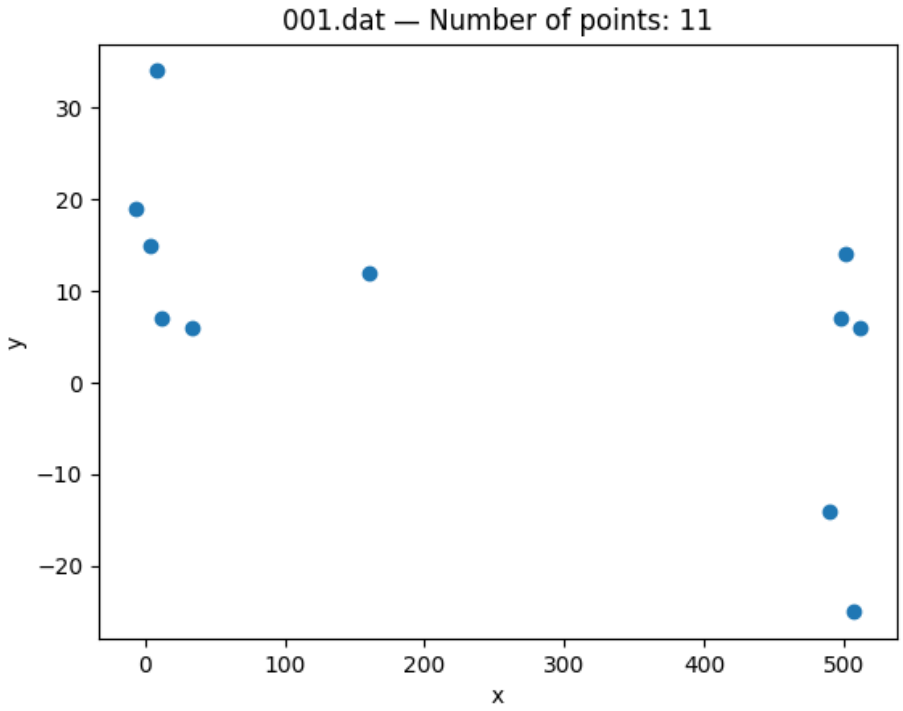
with open(FILE_PATH, "r") as file:
    n = int(file.readline())

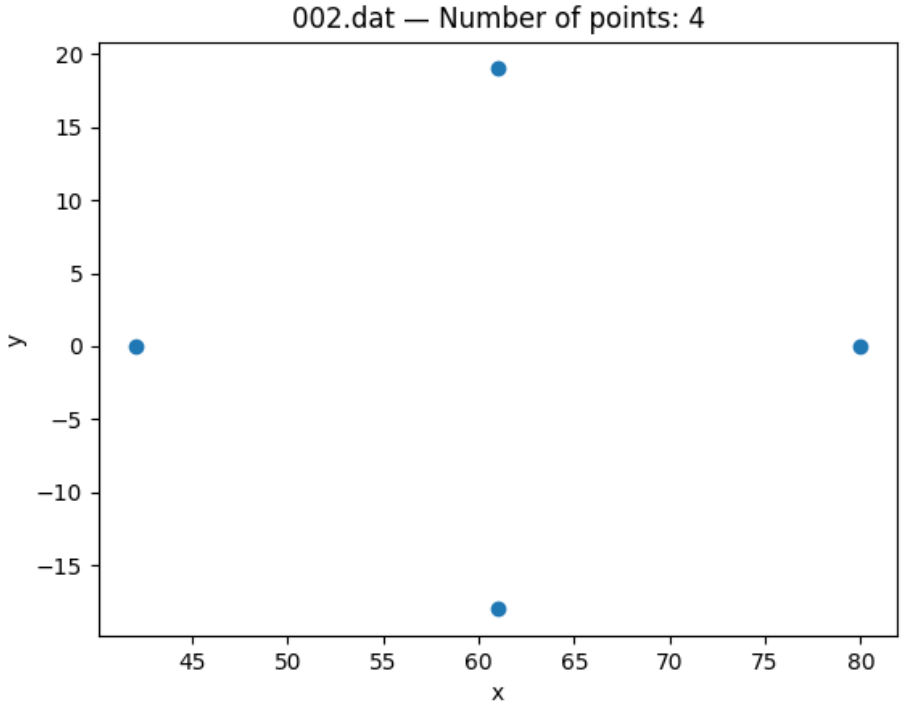
    x = []
    y = []

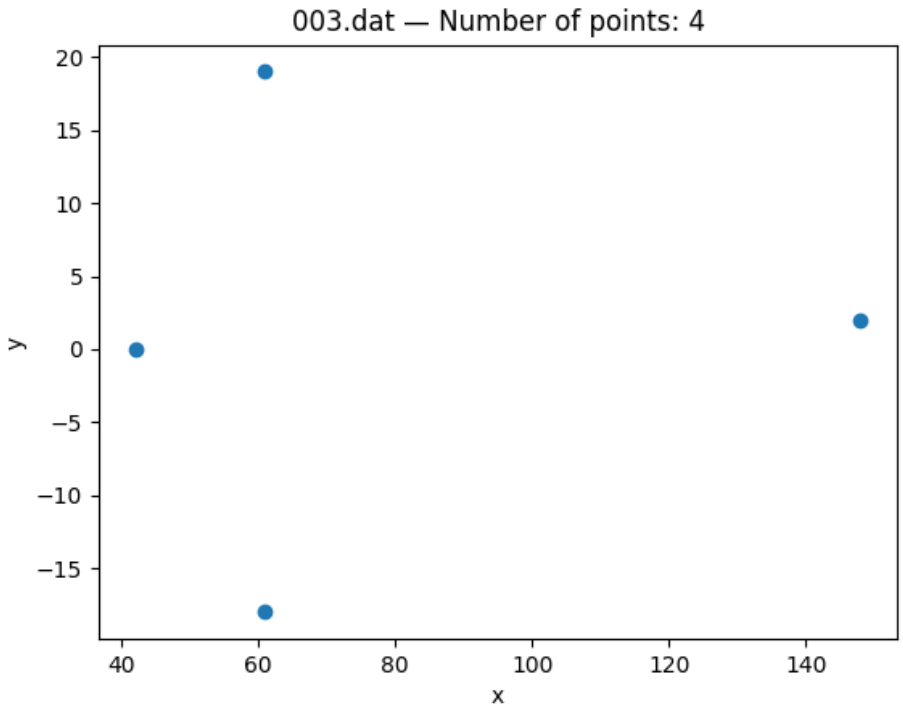
    for i in range(n):
        a, b = map(float, file.readline().split())
        x.append(a)
        y.append(b)

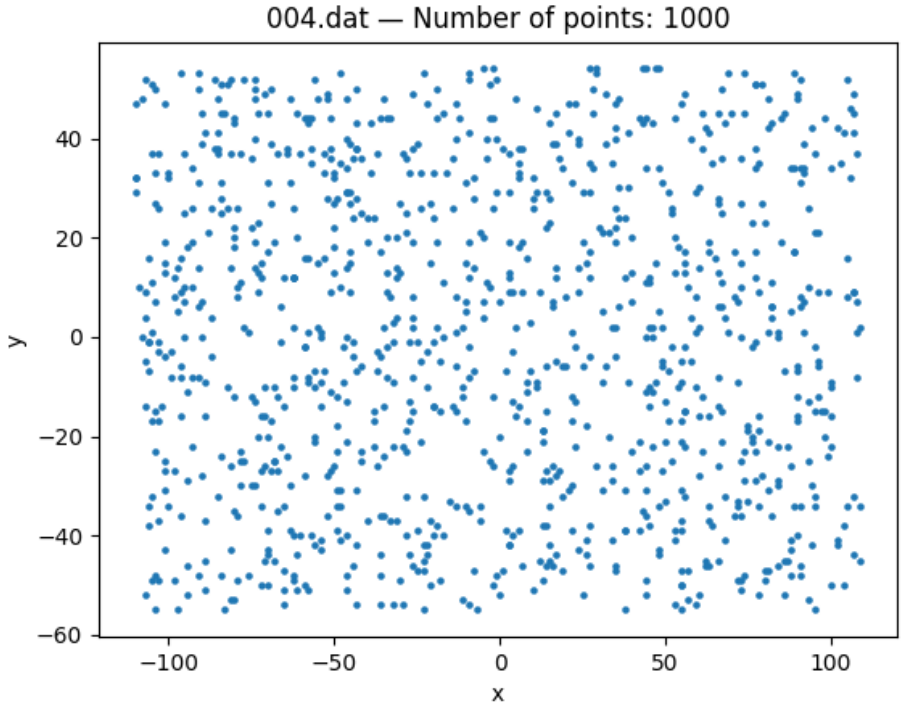
plt.scatter(x, y, s=5, label=FILE_NAME)
plt.xlabel('x')
plt.ylabel('y')
plt.title(f"{FILE_NAME} - Number of points: {n}")
plt.legend()
plt.show()
```

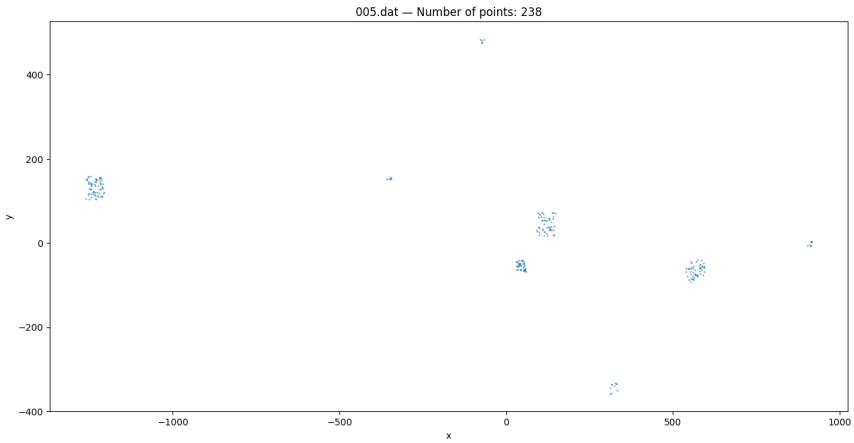
Результат


001.dat — Number of points: 11
30
20
10
0
−10
−20
y
0
100
200
300
400
500
x


002.dat — Number of points: 4
20
15
10
5
0
−5
−10
−15
y
45
50
55
60
65
70
75
80
x


003.dat — Number of points: 4
20
15
10
5
0
−5
−10
−15
y
40
60
80
100
120
140
x


004.dat — Number of points: 1000
40
20
0
−20
−40
−60
y
−100
−50
0
50
100
x


005.dat — Number of points: 238
400
200
0
−200
−400
y
−1000
−500
0
500
1000
x

Задание 2

Код программы

```
import matplotlib.pyplot as plt

with open(
    "C:/Users/user/Downloads/dead_moroz/Task_2/frames.dat",
    "r"
) as file:
    lines = file.readlines()

frames = []

for i in range(0, len(lines), 2):
    x = list(map(float, lines[i].split()))
    y = list(map(float, lines[i + 1].split()))
    frames.append((x, y))

fig, axes = plt.subplots(
    (len(frames) + 1) // 2, 2,
    figsize=(12, 8)
)

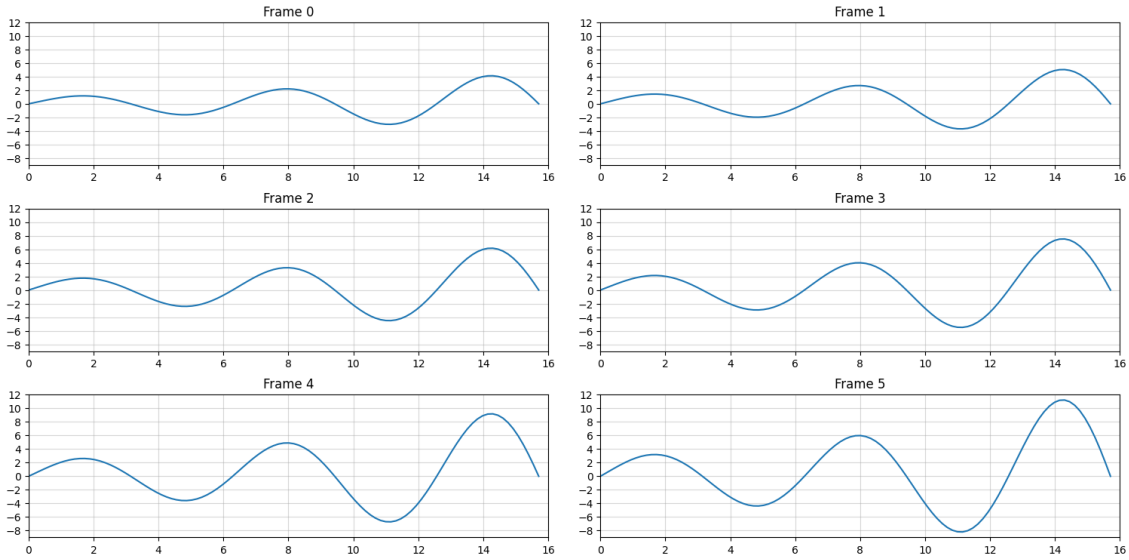
axes = axes.flatten()

for i in range(len(frames)):
    axes[i].plot(frames[i][0], frames[i][1])
    axes[i].set_title(f"Frame {i}")
    axes[i].set_xlim(0, 16)
    axes[i].set_ylim(-9, 12)
    axes[i].set_xticks(range(0, 17, 2))
    axes[i].set_yticks(range(-8, 13, 2))
    axes[i].grid(True, alpha=0.5)

for i in range(len(frames), len(axes)):
    axes[i].axis("off")

plt.tight_layout()
plt.show()
```

Результат



Задание 3

Код программы

```
import matplotlib.pyplot as plt

teachers = {}
groups = {}

with open(
    r"C:\Users\user\Downloads\dead_moroz\Task_3\students.txt",
    "r"
) as file:
    for line in file:
        teacher, group, grade = line.strip().split(";")
        grade = int(grade)

        if teacher not in teachers:
            teachers[teacher] = {}
        if grade not in teachers[teacher]:
            teachers[teacher][grade] = 0
        teachers[teacher][grade] += 1

        if group not in groups:
            groups[group] = {}
        if grade not in groups[group]:
            groups[group][grade] = 0
        groups[group][grade] += 1

grades = []

for data in teachers.values():
    for grade in data:
        if grade not in grades:
            grades.append(grade)

grades.sort()

fig, axes = plt.subplots(2, 1, figsize=(10, 10))

# Grades by preps
data = teachers
bar_numbers = range(len(data))
labels = list(data.keys())
bottom = [0] * len(data)

for grade in grades:
    values = [data[name].get(grade, 0) for name in labels]
    axes[0].bar(
        bar_numbers, values,
        bottom=bottom, label=str(grade)
    )
    bottom = [
        bottom[i] + values[i]
        for i in range(len(values))
    ]

axes[0].set_xticks(list(bar_numbers))
axes[0].set_xticklabels(labels)
axes[0].set_title("Grades by teacher")
```

```
axes[0].set_ylabel("Number of students")
axes[0].legend(title="Grade")
axes[0].grid(axis="y", alpha=0.3)

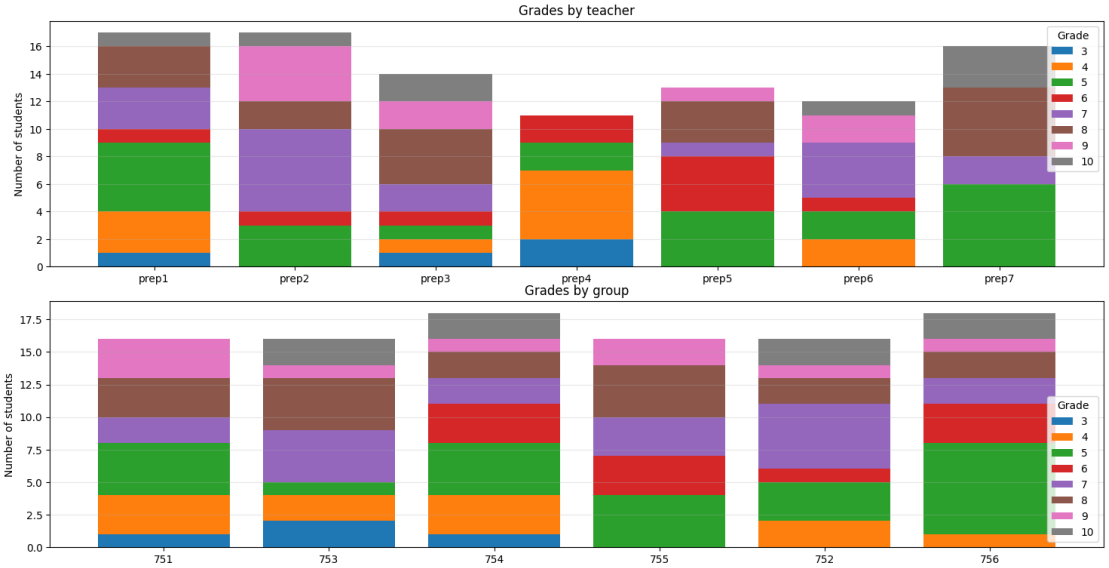
# Grades by group
data = groups
bar_numbers = range(len(data))
labels = list(data.keys())
bottom = [0] * len(data)

for grade in grades:
    values = [data[name].get(grade, 0) for name in labels]
    axes[1].bar(
        bar_numbers, values,
        bottom=bottom, label=str(grade)
    )
    bottom = [
        bottom[i] + values[i]
        for i in range(len(values))
    ]

axes[1].set_xticks(list(bar_numbers))
axes[1].set_xticklabels(labels)
axes[1].set_title("Grades by group")
axes[1].set_ylabel("Number of students")
axes[1].legend(title="Grade")
axes[1].grid(axis="y", alpha=0.3)

plt.tight_layout()
plt.show()
```

Результаты



Дополнительный анализ

Самый халявный преподаватель: prep2

Самый нехалявный преподаватель: prep4

Самая раздолбайская группа: 754